

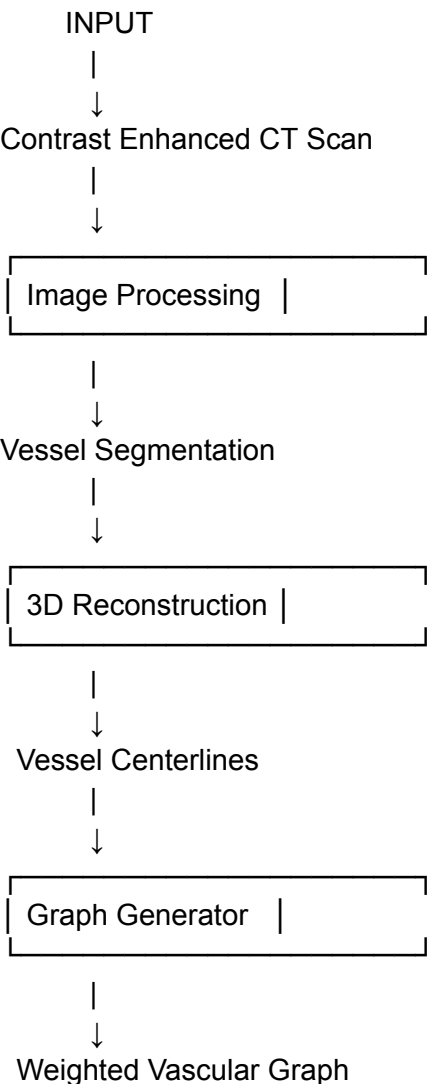
Computational Implementation Blueprint

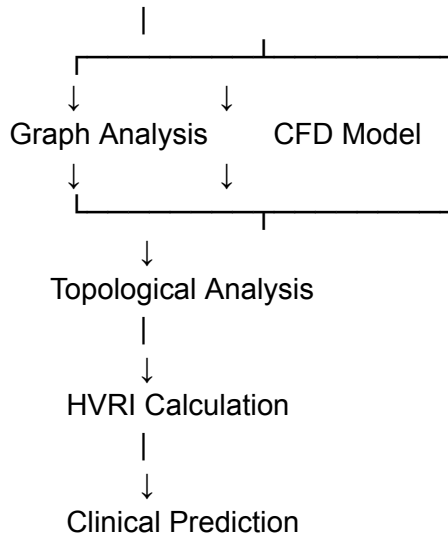
Project Name

Hepatic Vascular Resilience Digital Twin (HVR-DT)

1. Overall Software Architecture

The entire pipeline:





2. Technology Stack

Programming Language

Python

Reason:

- strongest biomedical ecosystem
- reproducible
- open-source

3. Environment Setup

Create:

hepatic-resilience/

Structure:

hepatic-resilience/

|

```
|— data/
|   |— raw_ct/
|   |— segmentation/
|   |— processed/
|
|— imaging/
|   |— preprocessing.py
|   |— segmentation.py
|
|— vascular/
|   |— centerlines.py
|   |— graph_builder.py
|
|— topology/
|   |— persistence.py
|   |— filtration.py
|
|— hemodynamics/
|   |— flow_model.py
|   |— resistance.py
|
|— resilience/
|   |— hvri.py
|   |— perturbation.py
|
|— analysis/
|   |— statistics.py
|   |— visualization.py
|
|— notebooks/
```

4. Phase 0 — Learn the Tools

Before touching liver data:

Build toy examples.

Install:

pip install numpy

```
pip install scipy
pip install matplotlib
pip install networkx
pip install gudhi
pip install ripser
pip install scikit-learn
pip install nibabel
pip install SimpleITK
```

5. Phase 1 — Generate Synthetic Vascular Networks

Before CT.

Why?

Because you need to prove your mathematics works.

Create:

Network A

Healthy vascular tree:



Expected:

High connectivity

High HVRI

Network B

Fragile network:

|
|
|
|

Expected:

Low HVRI

Experiment 1

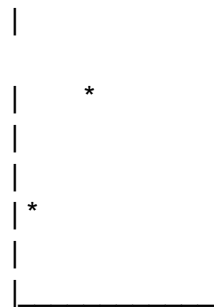
Question:

Does HVRI distinguish robust vs fragile networks?

Output:

Plot:

HVRI



Fragile Robust

6. Phase 2 — Import Real Vessel Data

Start with:

Dataset

3D-IRCADb

Workflow:

CT (.nii/.mha)

↓

Segmentation mask

↓

3D surface

Tools:

Medical image processing

SimpleITK

Example:

```
import SimpleITK as sitk
```

```
image=sitk.ReadImage(  
"patient001.nii"  
)
```

7. Phase 3 — Vessel Reconstruction

Goal:

Convert:

voxel image

↓

continuous vessel model

Tool:

VMTK

Pipeline:

Segmentation

↓

Surface extraction

↓

Centerline extraction

Output:

Example:

centerline.vtp

8. Phase 4 — Construct the Vascular Graph

Input:

centerline points

Output:

NetworkX graph:

```
import networkx as nx
```

```
G=nx.Graph()
```

Add nodes:

```
G.add_node(
1,
position=(x,y,z),
pressure=P
)
```

Add edges:

```
G.add_edge(
1,
2,
length=L,
radius=r
)
```

Final object:

$G=(V,E,W)$ $G=(V,E,W)$ $G=(V,E,W)$

9. Phase 5 — Hemodynamic Layer

Initially:

DO NOT start with full CFD.

Too expensive.

Start with reduced physics.

Resistance

Poiseuille equation:

$$R = \frac{8\mu L}{\pi r^4}$$

Python:

```
R=(8*mu*L)/(np.pi*r**4)
```

Flow approximation

Using:

$$Q = \Delta P R Q = \frac{\Delta P}{R} Q = R \Delta P$$

Store:

```
edge["flow"]=Q
```

10. Phase 6 — Spectral Analysis

Create weighted adjacency matrix.

Example:

```
A=nx.adjacency_matrix(  
G,  
weight="flow"  
)
```

Compute:

Laplacian

$$L = D - A = D - A = D - A$$

Python:

```
nx.laplacian_matrix(G)
```

Compute eigenvalues:

```
np.linalg.eigvals(L)
```

Extract:

$\lambda_2 \backslash \lambda_2$

11. Phase 7 — Persistent Homology

Use:

GUDHI

Input:

weighted graph

Filtration:

Example:

radius:

$f(e)=r, f(e)=r, f(e)=r$

Compute:

`import gudhi`

`complex=gudhi.SimplexTree()`

Output:

Persistence diagram:

Birth	Death
-------	-------

0.2	0.8
-----	-----

0.5	1.5
-----	-----

0.7 3.0

Features:

Calculate:

- lifetime
 - entropy
 - total persistence
-

12. Phase 8 — Perturbation Simulation

This is critical.

Simulate:

"what happens if a vessel is lost?"

Algorithm:

For each edge:

remove edge

↓

recalculate flow

↓

measure function loss

Pseudo:

for vessel in G.edges():

 G.remove_edge(vessel)

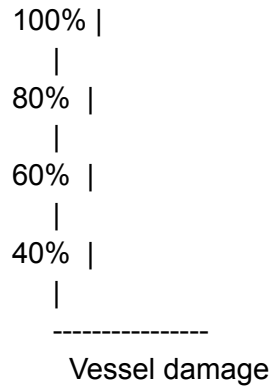
 loss=calculate_flow_loss()

store(loss)

Output:

Resilience curve:

Function



13. Phase 9 — HVRI Calculation

Implement:

HVRI =
w1*spectral_score
+
w2*topology_score
+
w3*entropy_score
+
w4*flow_score

Initially:

equal weights:

w1=w2=w3=w4=0.25

Later:

optimize statistically.

14. Visualization System

Important for surgeon presentation.

Create:

3D viewer

Show:

- liver
- vessels
- important branches

Tools:

- Plotly
 - VTK
-

Example visualization:

Color:

large importance vessels

highlight:

fragile regions

15. Machine Learning Module

Only AFTER features exist.

Dataset:

Rows:

patients

Columns:

features

Example:

Patient	HVRI	λ_2	Entropy	Outcome
1	0.82	0.3	1.4	Good
2	0.31	0.1	0.6	Poor

Models:

Start:

1. Logistic regression
 2. Random forest
 3. XGBoost
-

16. Validation Experiments

Experiment 1

Topology only.

Question:

Does topology distinguish networks?

Experiment 2

CFD only.

Question:

Does flow predict resilience?

Experiment 3

Topology + CFD.

Question:

Does combination outperform?